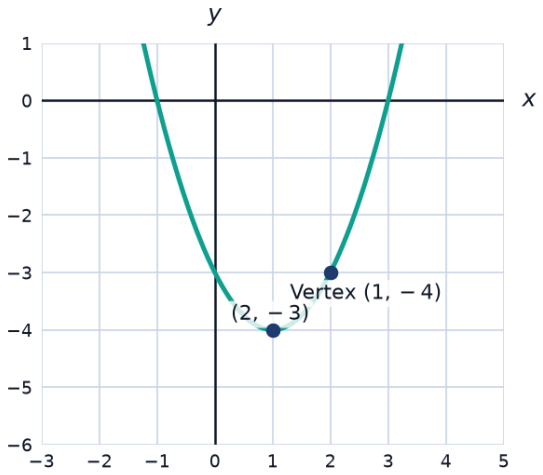
Quadratics & Nonlinear Equations

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Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

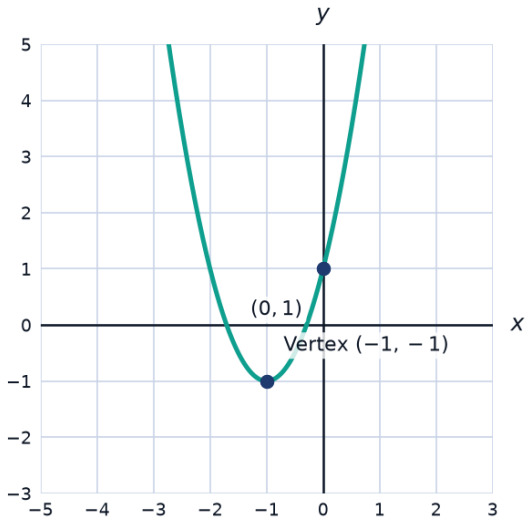
1 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the community theater arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



2 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

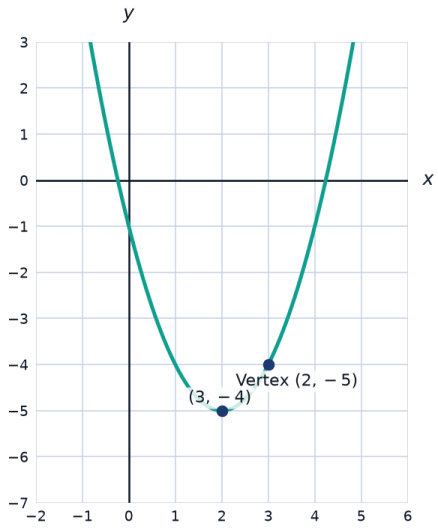
The graph of the quadratic function $y = q(x)$ for the mountain observatory arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



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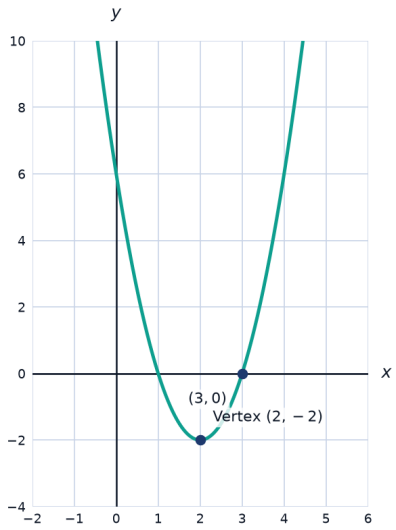
3 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the solar microgrid facility arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



4 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

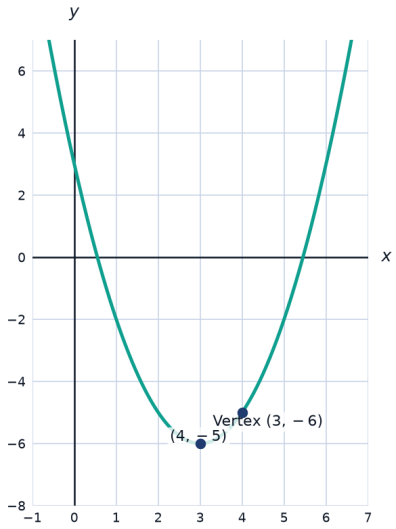
The graph of the quadratic function $y = q(x)$ for the publishing house arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



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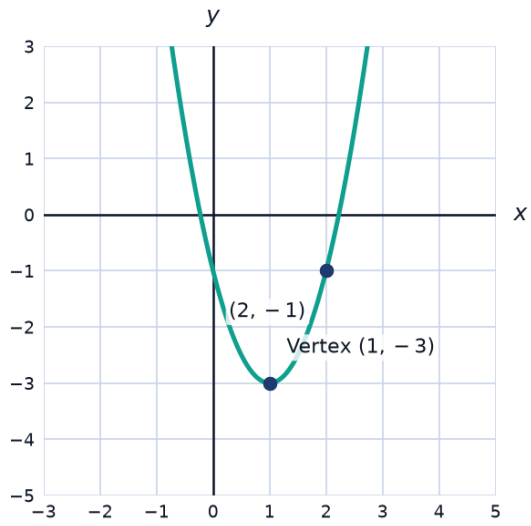
5 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the hydroponic greenhouse arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



6 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

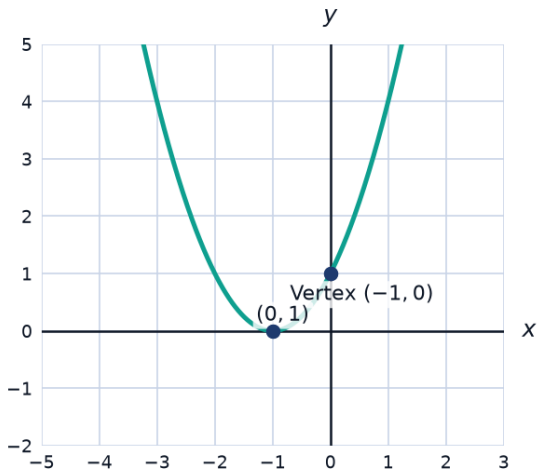
The graph of the quadratic function $y = q(x)$ for the robotics club arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



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7 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the wildlife sanctuary arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the robotics club, the quadratic function $h(u) = -u^2 + 28u - 140$ attains its maximum at $u = k$. What is the maximum value $h(k)$?

9 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), if $f(x) = x^2 - 6x - 2$, what is $f(5)$?

10 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the campus coffee roastery, for all $x \neq 5$, the rational expression $\frac{x^2 - 25}{x - 5}$ simplifies to $x + f$. What is the scaled constant $3f$?

11 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(x) = 2x^2 + 3x + 5$, what is $f(1)$?

12 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $f(x) = 5x^2 + 3$, what is $f(4)$?

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13 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 36), if $f(x) = x^2 - 8x + 6$, what is $f(5)$?

14 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, in a calculation for the community orchard (case 44), if $f(x) = 3x^2 - x + 7$, what is $f(-5)$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 68), what is the product of the solutions to $2x^2 + 6x - 20 = 0$?

16 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the botanical conservatory parabolic reflector equation $1x^2 - bx + 9 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

17 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x + y = 9$ and $x^2 - y^2 = 27$ in a calibration formula at the municipal library, what is the value of $x - y$?

18 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the product of the solutions to the quadratic equation $4x^2 - 7x + 20 = 0$ used in the community theater design model?

19 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the regional bakery parabolic reflector equation $1x^2 - bx + 36 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

20 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

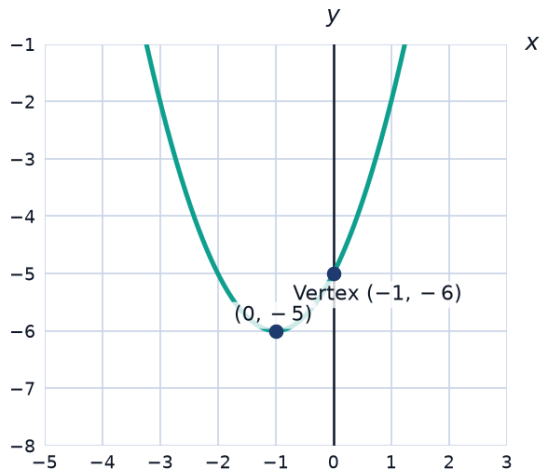
If $x + y = 12$ and $x^2 - y^2 = 72$ in a calibration formula at the coastal aquarium, what is the value of $x - y$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the botanical conservatory arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. -5

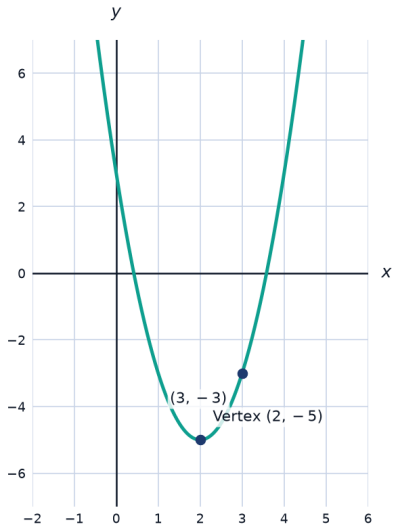
b. -3

c. -6

d. -4

22 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the municipal library arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. 5

b. -5

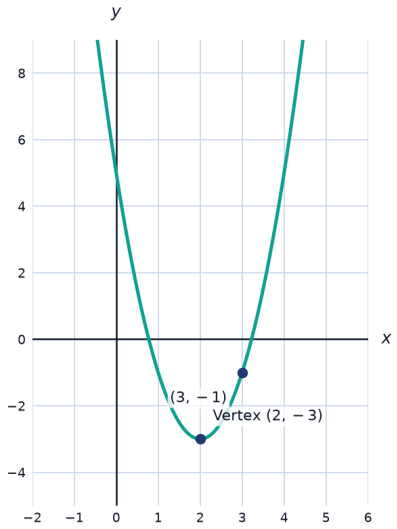
c. -1

d. 3

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23 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the regional bakery arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. 1

b. 5

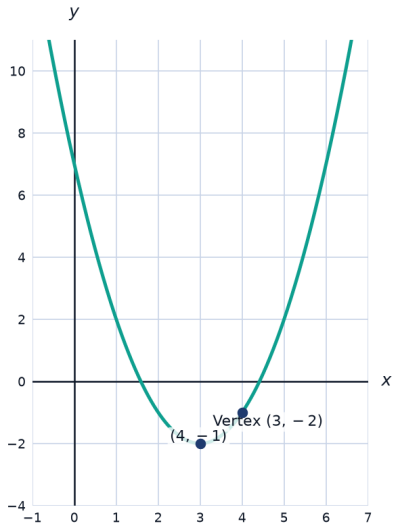
c. 7

d. −3

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24 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the coastal aquarium arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. 7

b. 9

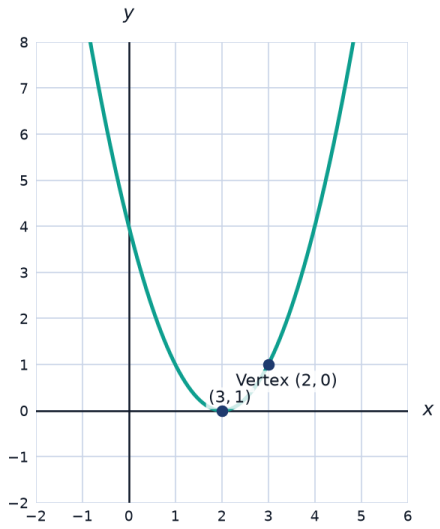
c. −2

d. 2

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25 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the science museum arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. 0

b. 3

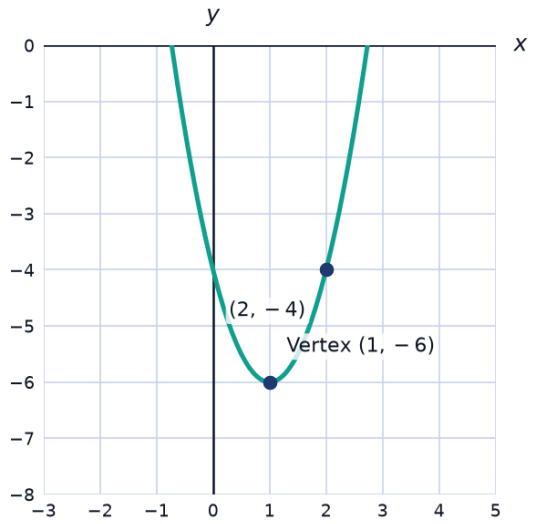
c. 4

d. 6

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26 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the urban bike-share station arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. -3

b. -4

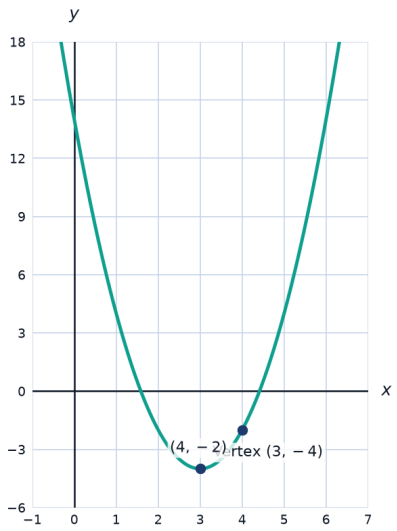
c. -2

d. -6

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27 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the ceramic studio arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. 16

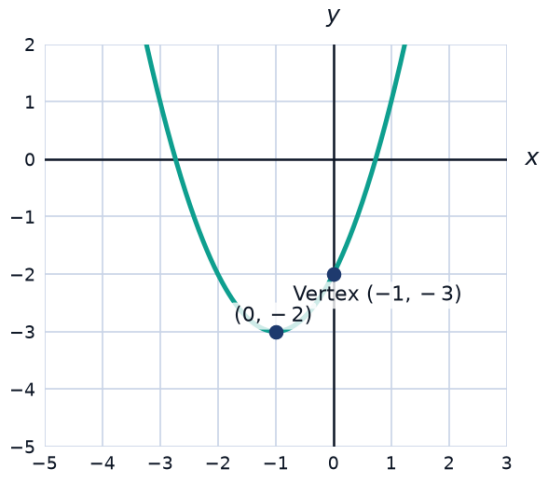
b. -4

c. 1

d. 14

28 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the agricultural cooperative arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?



a. -3

b. -1

c. -2

d. 0

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Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas**

In a model for the art gallery, consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. What is the remainder $f(2)$ upon division by $(x - 2)$?

a. -22

b. -12

c. -7

d. -10

30 **[Intermediate]** • **[MCQ]** • **Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas**

In a model for the culinary institute, the exponential equation $6^{2x-1} = 6^{4+5}$ holds for real x. What is the quantity $f = 2x$?

a. 14

b. 5

c. 10

d. 16

31 **[Intermediate]** • **[MCQ]** • **Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas**

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), what is the vertex of the parabola $y = x^2 - 14x + 53$?

a. (7, -4)

b. (-7, 4)

c. (7, 4)

d. (8, 4)

32 **[Intermediate]** • **[MCQ]** • **Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas**

In a model for the sports complex, the radical function $f(x) = \sqrt{7x + 11}$ is defined for $7x + 11 \geq 0$. What is the product $7x$ when $f(x) = 9$?

a. 80

b. 59

c. 70

d. 77

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33 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the maple syrup farm, in a calculation for the honeybee apiary (case 16), how many x-intercepts does the graph of $y = -2x^2 - 5x - 3$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

34 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), how many x-intercepts does the graph of $y = -x^2 - x + 2$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

35 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), how many x-intercepts does the graph of $y = x^2 - 2x + 4$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

36 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), if $f(x) = 2x^2 + 2x - 2$, what is $f(5)$?

a. 63

b. 58

c. 49

d. 66

37 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the national park visitor center, in a calculation for the environmental club (case 21), if $f(x) = 4x^2 + 2x + 7$, what is $f(-4)$?

a. 63

b. 66

c. 57

d. 62

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the maple syrup farm, for what sum of x-values does the graph of $y = x^2 + x - 12$ cross the x-axis?

a. -6

b. -1

c. 3

d. 2

39 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the planetarium, in a calculation for the carpentry workshop (case 67), what is the sum of the solutions to $2x^2 - 4x - 70 = 0$?

a. 1

b. -7

c. 2

d. -3

40 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, the quadratic graph $p(t) = 6t^2 + qt + 6$ is tangent to the horizontal axis with $q > 0$. What is the positive coefficient q?

a. 13

b. 18

c. 12

d. 3

41 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the marine research vessel, for what sum of x-values does the graph of $y = -x^2 + 4x + 12$ cross the x-axis?

a. 7

b. 5

c. 11

d. 4

42 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the university lecture hall parabolic reflector equation $1x^2 - bx + 49 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

a. 49

b. 18

c. 14

d. 7

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43 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x + y = 17$ and $x^2 - y^2 = 102$ in a calibration formula at the culinary institute, what is the value of $x - y$?

a. 85

b. 6

c. 11

d. 8

44 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the product of the solutions to the quadratic equation $2x^2 - 25x + 14 = 0$ used in the wildlife sanctuary design model?

a. 7

b. 14

c. 25

d. 9

45 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the water treatment plant parabolic reflector equation $1x^2 - bx + 9 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

a. 3

b. 9

c. 10

d. 6

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